CARA 2.0 Kinematics

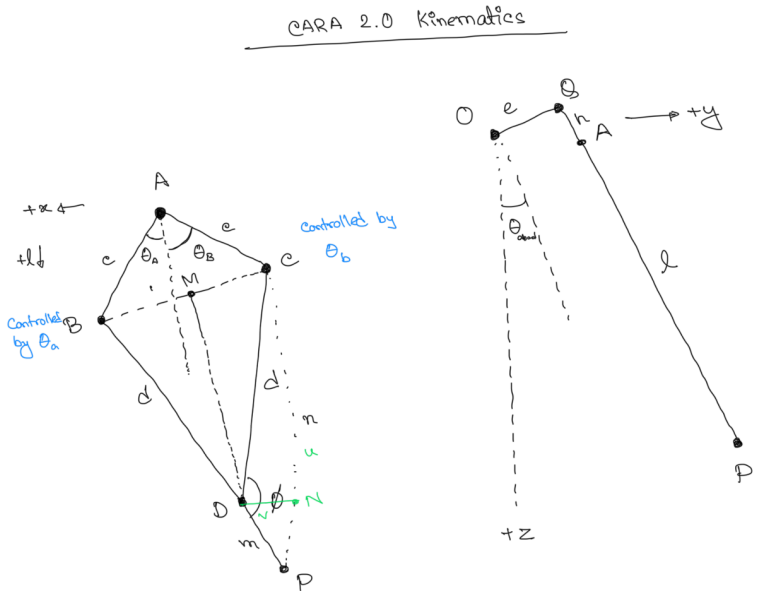
IK, FK and Jacobians

Typed and checked from the handwritten notes supplied on 19 October 2026. Diagrams are reproduced or redrawn for readability. Corrections and branch assumptions are stated explicitly.

1 IK

Geometry and notation

The target is the foot centre $\mathbf{p} = [x, y, z]^T$. The planar linkage has its origin at A; the abduction origin is O. The downward planar foot coordinate is ℓ, and $L = h + \ell$ includes the mounting offset. Write $t = \theta_{\text{abad}}$ to avoid confusing the abduction angle with θ_a.



Original geometry sketches from the handwritten document.

Symbol	Meaning
c	$AB = AC$, proximal length
d	$BD = CD$, distal length
m	DP, foot-centre offset in the linkage plane
ϕ	internal angle $\angle CDP$, with rotation sense below
e	fixed sideways offset
h	fixed downward offset

Angles are in radians in calculations. The planar coordinates are ordered $[x, \ell]^T$, with $+\ell$ downward and $+x$ as shown in the original sketch. The formulas define the axis signs; do not substitute encoder readings without calibrating their offsets and directions.

Recover the abduction angle and the planar target

From the mounting geometry,

$$y = e\cos t + L\sin t, \tag{1.1}$$
$$z = -e\sin t + L\cos t, \qquad L = h + \ell. \tag{1.2}$$

Squaring and adding cancels the cross terms:

$$y^2 + z^2 = e^2(\cos^2 t + \sin^2 t) + L^2(\sin^2 t + \cos^2 t) \tag{1.3}$$
$$= e^2 + L^2. \tag{1.4}$$

Therefore the two algebraic radial choices are

$$L = \eta\sqrt{y^2 + z^2 - e^2}, \qquad \eta \in \{-1, +1\}, \qquad \ell = L - h. \tag{1.5}$$

The usual downward leg branch takes $\eta = +1$. A negative L is a different geometric branch; it must not be silently discarded if it is physically permitted.

Multiply the first mounting equation by L and subtract e times the second:

$$Ly - ez = (L^2 + e^2)\sin t. \tag{1.6}$$

Similarly,

$$ey + Lz = (L^2 + e^2)\cos t. \tag{1.7}$$

Their common positive denominator is unnecessary in atan2:

$$\boxed{t = \operatorname{atan2}\big(Ly - ez,\ ey + Lz\big).} \tag{1.8}$$

Here $\operatorname{atan2}(u, v)$ means the angle whose sine is proportional to u and cosine to v, as in Python's `atan2(u,v)`.

Condition. $y^2 + z^2 \geq e^2$ is required for a real L. Also $L^2 + e^2 > 0$ is needed to identify t from this target. If $L = e = 0$, then $y = z = 0$ and the target does not determine the abduction angle.

At equality $y^2 + z^2 = e^2$, the two radial branches meet. Keep this boundary separate from the regular interior of the workspace.

The remaining IK problem is now planar: find θ_b, D, and θ_a for the target $P = (x, \ell)$.

Recover θ_b from the fixed distance CP

Because C, D, P form a rigid triangle, the cosine rule gives

$$n = CP = \sqrt{d^2 + m^2 - 2dm\cos\phi}. \tag{1.9}$$

With A as the planar origin,

$$C = \begin{bmatrix} -c\sin\theta_b \\ c\cos\theta_b \end{bmatrix}, \qquad (x - x_C)^2 + (\ell - \ell_C)^2 = n^2. \tag{1.10}$$

Expanding the distance constraint,

$$(x + c\sin\theta_b)^2 + (\ell - c\cos\theta_b)^2 = n^2, \tag{1.11}$$

$$x^2 + \ell^2 + c^2 + 2cx\sin\theta_b - 2c\ell\cos\theta_b = n^2. \tag{1.12}$$

Define

$$f^2 = x^2 + \ell^2, \qquad k = \frac{x^2 + \ell^2 + c^2 - n^2}{2c}. \tag{1.13}$$

Then

$$\ell\cos\theta_b - x\sin\theta_b = k. \tag{1.14}$$

Introduce the perpendicular combination $\bar{k} = x\cos\theta_b + \ell\sin\theta_b$. Squaring both combinations gives

$$k^2 + \bar{k}^2 = (x^2 + \ell^2)(\sin^2\theta_b + \cos^2\theta_b) = f^2, \quad \bar{k} = \epsilon_b\sqrt{f^2 - k^2}. \tag{1.15}$$

Solving these two linear equations for sine and cosine,

$$\sin\theta_b = \frac{\ell\bar{k} - xk}{f^2}, \qquad \cos\theta_b = \frac{\ell k + x\bar{k}}{f^2}. \tag{1.16}$$

Thus both candidate branches are

$$\boxed{\theta_b = \operatorname{atan2}(\ell\bar{k} - xk,\ \ell k + x\bar{k}), \quad \epsilon_b = \pm 1.} \tag{1.17}$$

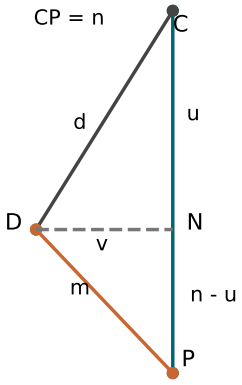
Condition. $c > 0$, $f > 0$, and $f^2 - k^2 \geq 0$. Equivalently, $|c - n| \leq f \leq c + n$. At $f = 0$, this formula is undefined: a solution requires $c = n$, and θ_b is not determined by CP alone.

For the usual branch, the same equation is $\theta_b = \arccos(k/f) - \operatorname{atan2}(x, \ell)$. This explains the **minus sign**; the plus sign in the earlier seven-page PDF summary is a typo.

Recover the distal joint D

Once a candidate θ_b is known, C is known. The point D must lie on both circles:

$$\|D - C\| = d, \qquad \|D - P\| = m. \tag{1.18}$$



Redrawn triangle used to recover D from C and P (schematic, not to scale).

Set $\Delta_x = x - x_C$, $\Delta_\ell = \ell - \ell_C$, so $\Delta_x^2 + \Delta_\ell^2 = n^2$. Let $u = CN$ and $v = DN$. From the two right triangles,

$$u^2 + v^2 = d^2, \tag{1.19}$$
$$(n - u)^2 + v^2 = m^2. \tag{1.20}$$

Subtracting eliminates v^2:

$$n^2 - 2nu = m^2 - d^2, \qquad u = \frac{n^2 + d^2 - m^2}{2n}, \quad v = \sqrt{d^2 - u^2}. \tag{1.21}$$

A displacement along CP is proportional to $[\Delta_x, \Delta_\ell]^T$; a perpendicular displacement is proportional to $[\Delta_\ell, -\Delta_x]^T$. Hence

$$\boxed{\begin{aligned} x_D &= x_C + \frac{u}{n}\Delta_x + \sigma\frac{v}{n}\Delta_\ell, \\ \ell_D &= \ell_C + \frac{u}{n}\Delta_\ell - \sigma\frac{v}{n}\Delta_x, \qquad \sigma = \pm 1. \end{aligned}} \tag{1.22}$$

Condition. $n > 0$ and $d^2 - u^2 \geq 0$, equivalently $|d - m| \leq n \leq d + m$. If $n = 0$, $C = P$ and this construction must be replaced by its degenerate geometry.

The two signs reflect D across line CP. Retain only the orientation consistent with the rigid foot relation in the FK section, not merely the circle lengths.

Recover θ_a and select a consistent solution

The other proximal endpoint is

$$B = \begin{bmatrix} c\sin\theta_a \\ c\cos\theta_a \end{bmatrix}, \qquad (x_D - c\sin\theta_a)^2 + (\ell_D - c\cos\theta_a)^2 = d^2. \tag{1.23}$$

Expanding gives

$$x_D\sin\theta_a + \ell_D\cos\theta_a = \frac{x_D^2 + \ell_D^2 + c^2 - d^2}{2c} = k_a. \tag{1.24}$$

Define the perpendicular combination and its two signs:

$$\bar{k}_a = \ell_D\sin\theta_a - x_D\cos\theta_a = \epsilon_a\sqrt{x_D^2 + \ell_D^2 - k_a^2}, \quad \epsilon_a = \pm 1. \tag{1.25}$$

Therefore

$$\sin\theta_a = \frac{x_D k_a + \ell_D\bar{k}_a}{x_D^2 + \ell_D^2}, \tag{1.26}$$

$$\cos\theta_a = \frac{\ell_D k_a - x_D\bar{k}_a}{x_D^2 + \ell_D^2}, \tag{1.27}$$

$$\theta_a = \operatorname{atan2}(x_D k_a + \ell_D\bar{k}_a,\ \ell_D k_a - x_D\bar{k}_a). \tag{1.28}$$

Condition. $x_D^2 + \ell_D^2 > 0$ and $x_D^2 + \ell_D^2 - k_a^2 \geq 0$. Equivalently, $|c - d| \leq AD \leq c + d$. If $D = A$, the distance constraint admits any θ_a only when $c = d$.

IK branch and acceptance conditions

Enumerate the allowed signs $\eta, \epsilon_b, \sigma, \epsilon_a$. For every candidate:

1. Check the real-root and nonzero-denominator conditions above.
2. Check all four link lengths and the signed rigid-foot relation

$$P - D = \frac{m}{d}\begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix}(C - D).$$

3. Apply joint limits and keep the intended assembly branch. During motion, choose a continuous solution close to the previous angles.
4. Evaluate FK and accept only if $\|\mathrm{FK}(\mathbf{q}) - \mathbf{p}_{\mathrm{target}}\|$ is below the chosen tolerance.

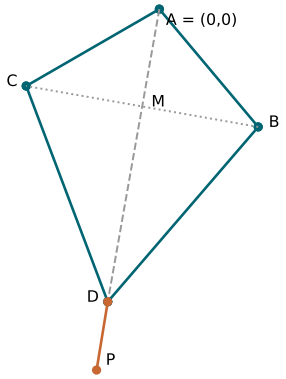
Different algebraic sign combinations are not automatically different physical solutions. Some are duplicates or violate the required foot orientation. Do not repair an unreachable target by broadly clipping a negative radicand; only tiny roundoff violations may be tolerated.

2 FK

Find B, C, then the diagonal AD

The inputs are $\mathbf{q} = [\theta_a, \theta_b, t]^T$. The proximal endpoints are

$$B = \begin{bmatrix} c\sin\theta_a \\ c\cos\theta_a \end{bmatrix}, \qquad C = \begin{bmatrix} -c\sin\theta_b \\ c\cos\theta_b \end{bmatrix}. \tag{2.1}$$



Redrawn linkage. M is the midpoint of BC; AD bisects BC.

The equal distal lengths imply

$$\|D - B\|^2 = d^2, \qquad \|D - C\|^2 = d^2, \tag{2.2}$$

$$(x_B - x_C)x_D + (\ell_B - \ell_C)\ell_D = 0. \tag{2.3}$$

Using sum-to-product identities, with

$$s = \frac{\theta_a + \theta_b}{2}, \qquad \delta = \frac{\theta_a - \theta_b}{2}, \tag{2.4}$$

this becomes $2c\sin s\,(x_D\cos\delta - \ell_D\sin\delta) = 0$. For $\sin s \neq 0$, write

$$x_D = \rho\sin\delta, \qquad \ell_D = \rho\cos\delta. \tag{2.5}$$

Substituting in $\|D - B\|^2 = d^2$ yields

$$\rho^2 + c^2 - 2\rho c\cos(\delta - \theta_a) = d^2 \quad \Longrightarrow \quad \rho^2 - 2c\cos s\,\rho + c^2 - d^2 = 0. \tag{2.6}$$

Therefore

$$\boxed{\rho = c\cos s + \tau\sqrt{d^2 - c^2\sin^2 s}, \quad \tau = \pm 1.} \tag{2.7}$$

ρ is a signed diagonal coordinate; on the tested lower branch $\tau = +1$ and $\rho = AD > 0$.

FK conditions and a geometric construction

For real intersections,

$$T^2 = d^2 - c^2 \sin^2 s \geq 0. \tag{2.8}$$

If $T = 0$, the two circle intersections meet. For a regular differentiable configuration, require $T > 0$ and retain a fixed branch.

If $\sin s = 0$, then $B = C$. The two distal circle constraints are identical and do not determine a unique D. The radial formula may return a limiting point, but it must not be presented as a unique FK solution at that degenerate configuration.

An equivalent construction, useful for checking the radial solution, is

$$\mathbf{v} = C - B, \qquad b = \|\mathbf{v}\|, \qquad M = \frac{B + C}{2}, \tag{2.9}$$

$$D = M + \beta \sqrt{d^2 - \frac{b^2}{4}} \, \frac{1}{b} \begin{bmatrix} -v_\ell \\ v_x \end{bmatrix}, \qquad \beta = \pm 1. \tag{2.10}$$

This requires $0 < b \leq 2d$. The sign β must be matched to the chosen τ; its meaning changes with the ordering and direction of $C - B$.

Extend D to the foot centre P

The vector $D \to C$ has length d. Rotate it in the signed sense shown below and scale it to the foot-offset length m:

$$R_\phi = \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix}, \qquad P = D + \frac{m}{d} R_\phi (C - D). \tag{2.11}$$

In components,

$$\boxed{\begin{aligned} x &= x_D + \frac{m}{d}\big[(x_C - x_D)\cos\phi + (\ell_C - \ell_D)\sin\phi\big], \\ \ell &= \ell_D + \frac{m}{d}\big[-(x_C - x_D)\sin\phi + (\ell_C - \ell_D)\cos\phi\big]. \end{aligned}} \tag{2.12}$$

The sign of ϕ fixes the handedness of the rigid triangle. An internal angle alone does not distinguish a triangle from its reflection; the rotation matrix supplies that information.

Both branches influence P: C appears explicitly, while B acts through the location of D. There is no missing B term in the rigid offset equation.

Add the mounting offsets and abduction

At zero abduction, the foot position relative to O is

$$\mathbf{p}_0 = \begin{bmatrix} x \\ e \\ h+\ell \end{bmatrix}. \tag{2.13}$$

In the sign convention of the handwritten notes,

$$\boxed{\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{bmatrix} \begin{bmatrix} x \\ e \\ h+\ell \end{bmatrix}}. \tag{2.14}$$

Thus

$$y = e\cos t + (h+\ell)\sin t, \qquad z = -e\sin t + (h+\ell)\cos t. \tag{2.15}$$

In a standard right-handed Cartesian frame, this matrix is $R_x(-t)$. Preserve the specified convention when mapping it to CAD, ROS, or a motor encoder; the notes' sign convention is the definition used throughout this document.

Symbolic FK evaluation sequence

For any specified geometry and admissible joint angles, evaluate the equations in this order:

1. Calculate B, C, the half-sum s, and the half-difference δ.
2. Check the intersection conditions and select a fixed assembly branch τ.
3. Calculate ρ, then x_D and ℓ_D.
4. Apply the rigid foot offset to obtain x and ℓ.
5. Set $L = h + \ell$ and apply the abduction transformation to obtain y, z.

All lengths must use the same units. The output position uses those same units. The foot point P is the point specified by the rigid offset m, ϕ; it should not be confused with a ground-contact point unless the chosen geometry explicitly defines it that way.

The offsets e, h and all angle zero positions are geometric inputs to be supplied for the mechanism being studied. Coordinate-frame and encoder conventions must be defined consistently before using the resulting positions.

3 Jacobians

Define the velocity mapping

The translational Jacobian maps the three independent joint rates to foot-centre velocity:

$$\dot{\mathbf{p}} = J(\mathbf{q})\dot{\mathbf{q}}, \qquad J = \frac{\partial(x, y, z)}{\partial(\theta_a, \theta_b, t)}. \tag{3.1}$$

The planar quantities x, ℓ depend on θ_a, θ_b, but not on t. Differentiate the mounting equations first:

$$x_t = 0, \quad y_t = -e\sin t + (h + \ell)\cos t = z, \quad z_t = -e\cos t - (h + \ell)\sin t = -y. \tag{3.2}$$

For $i \in \{a, b\}$,

$$y_i = \sin t\, \ell_i, \qquad z_i = \cos t\, \ell_i, \tag{3.3}$$

where a subscript i means differentiation with respect to θ_i. Therefore

$$\boxed{J = \begin{bmatrix} x_a & x_b & 0 \\ \sin t\, \ell_a & \sin t\, \ell_b & z \\ \cos t\, \ell_a & \cos t\, \ell_b & -y \end{bmatrix}}. \tag{3.4}$$

The remaining task is to calculate x_a, x_b, ℓ_a, ℓ_b.

Differentiate the diagonal coordinate ρ

On a fixed branch,

$$s = \frac{\theta_a + \theta_b}{2}, \quad T = \sqrt{d^2 - c^2 \sin^2 s}, \quad \rho = c\cos s + \tau T. \tag{3.5}$$

Since $s_a = s_b = 1/2$, the chain rule gives

$$\frac{dT}{ds} = \frac{-2c^2 \sin s \cos s}{2T} = -\frac{c^2 \sin s \cos s}{T}. \tag{3.6}$$

Hence

$$\boxed{\rho_a = \rho_b = -\frac{c}{2}\sin s - \tau\frac{c^2 \sin s \cos s}{2T}}. \tag{3.7}$$

Correction to the handwriting: the square-root contribution has a minus sign. On the tested branch $\tau = +1$, both terms above are negative for $0 < s < \pi/2$. These derivatives require $T > 0$.

Differentiate the distal joint D

Let $r' = \rho_a = \rho_b$ and recall

$$\delta = \frac{\theta_a - \theta_b}{2}, \quad \delta_a = \frac{1}{2}, \quad \delta_b = -\frac{1}{2}, \qquad x_D = \rho \sin\delta, \quad \ell_D = \rho \cos\delta. \tag{3.8}$$

Using the product rule,

$$x_{D,a} = r' \sin\delta + \frac{\rho}{2}\cos\delta, \tag{3.9}$$

$$x_{D,b} = r' \sin\delta - \frac{\rho}{2}\cos\delta, \tag{3.10}$$

$$\ell_{D,a} = r' \cos\delta - \frac{\rho}{2}\sin\delta, \tag{3.11}$$

$$\ell_{D,b} = r' \cos\delta + \frac{\rho}{2}\sin\delta. \tag{3.12}$$

Thus the planar Jacobian of D is

$$J_D = \begin{bmatrix} r'\sin\delta + \frac{\rho}{2}\cos\delta & r'\sin\delta - \frac{\rho}{2}\cos\delta \\ r'\cos\delta - \frac{\rho}{2}\sin\delta & r'\cos\delta + \frac{\rho}{2}\sin\delta \end{bmatrix}. \tag{3.13}$$

Differentiate C and the rigid foot offset

Because C depends only on θ_b,

$$J_C = \begin{bmatrix} 0 & -c\cos\theta_b \\ 0 & -c\sin\theta_b \end{bmatrix}. \tag{3.14}$$

Set $\kappa = m/d$. Differentiating $P = D + \kappa R_\phi (C - D)$ gives

$$\boxed{J_P = (I - \kappa R_\phi) J_D + \kappa R_\phi J_C = \begin{bmatrix} x_a & x_b \\ \ell_a & \ell_b \end{bmatrix}}. \tag{3.15}$$

For example, the first column is

$$x_a = (1 - \kappa\cos\phi) x_{D,a} - \kappa \sin\phi\, \ell_{D,a}, \tag{3.16}$$

$$\ell_a = \kappa \sin\phi\, x_{D,a} + (1 - \kappa\cos\phi)\ell_{D,a}. \tag{3.17}$$

This shows why $C_a = 0$ does not remove the influence of motor a: motor a still moves D through B.

Complete the foot derivatives and the full Jacobian

The second column includes the nonzero derivative of C:

$$x_b = (1 - \kappa \cos\phi)x_{D,b} - \kappa \sin\phi\, \ell_{D,b} - \kappa c \cos(\phi - \theta_b), \tag{3.18}$$

$$\ell_b = \kappa \sin\phi\, x_{D,b} + (1 - \kappa \cos\phi)\ell_{D,b} + \kappa c \sin(\phi - \theta_b). \tag{3.19}$$

For instance, the last term in x_b follows from

$$\kappa(\cos\phi\, x_{C,b} + \sin\phi\, \ell_{C,b}) = -\kappa c(\cos\phi\cos\theta_b + \sin\phi\sin\theta_b).$$

Together with the first-column formulas on the previous page, these give every entry of

$$\boxed{J = \begin{bmatrix} x_a & x_b & 0 \\ \sin t\, \ell_a & \sin t\, \ell_b & z \\ \cos t\, \ell_a & \cos t\, \ell_b & -y \end{bmatrix}}. \tag{3.20}$$

If only the abduction joint moves, the velocity mapping reduces to

$$\dot{x} = 0, \qquad \dot{y} = z\dot{t}, \qquad \dot{z} = -y\dot{t}. \tag{3.21}$$

If lengths are expressed in a chosen length unit and joint angles in radians, the entries of J have units of length per radian.

An independent way to obtain J_D

Differentiating the two loop constraints directly gives

$$\underbrace{\begin{bmatrix} (D-B)^T \\ (D-C)^T \end{bmatrix}}_{A_D} J_D = \begin{bmatrix} (D-B)^T B_a & 0 \\ 0 & (D-C)^T C_b \end{bmatrix}, \tag{3.22}$$

where $B_a = c[\cos\theta_a, -\sin\theta_a]^T$ and $C_b = c[-\cos\theta_b, -\sin\theta_b]^T$. Solve this 2×2 system when $\det A_D \neq 0$. It provides the same derivatives without differentiating the radial formula.

Conditions, singularities and verification

The local Jacobian assumes a fixed assembly branch, differentiable FK, and angles in radians. It is a 3×3 *position* Jacobian, not a six-row position-and-orientation Jacobian.

Loop degeneracy. If $B = C$, the distal position is not unique. If $\det A_D = 0$, the vectors $D - B$ and $D - C$ are parallel and the loop equations do not give a regular local mapping from motor angles to D. The $T = 0$ circle-tangency case is one such boundary.

Loss of foot-velocity directions. Where a finite Jacobian exists, expanding its determinant gives

$$\det J = -(h + \ell)\,(x_a \ell_b - x_b \ell_a) = -L \det J_P. \tag{3.23}$$

Thus $L = 0$ or $\det J_P = 0$ causes a rank loss. Near these configurations, solving for joint rates can become sensitive. A nonzero determinant alone is not a measure of how far the mechanism is from singularity; inspect the smallest singular value or condition number as well.

Numerical derivative check. For a small angle ε and the unit vector $\mathbf{e}_i$ for joint i,

$$J_{\text{num},i} = \frac{\mathrm{FK}(\mathbf{q} + \varepsilon \mathbf{e}_i) - \mathrm{FK}(\mathbf{q} - \varepsilon \mathbf{e}_i)}{2\varepsilon}. \tag{3.24}$$

Compare the analytical and numerical Jacobians at several regular configurations while keeping the same assembly branch. Repeat the comparison for several small values of ε: an excessively small perturbation can amplify floating-point rounding, while a larger perturbation increases finite-difference approximation error.

Reusable verification checks. For any chosen geometry, check that the computed points preserve the link lengths, that the rigid-foot relation is satisfied, and that IK followed by FK returns the requested target. These checks establish mathematical consistency within the tested configurations; they do not replace joint-limit, collision, encoder-calibration, or physical testing.

Source basis: CARA 2.0 Kinematic'S.pdf (handwritten notes) and the earlier coaxial five-bar linkage derivation notes (sign comparison). This document retains symbolic derivations and conditions without dimensions or numerical results from a particular CAD assembly.